

ARTIFICIAL INTELLIGENCE LAB (IT-515)

Objectives:

Dated: 1 September,2026

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To study and design the heuristic algorithms

To implement heuristic algorithm on real life applications

1.Implement Forward and Backward Chaining for a given knowledge base and compare the reasoning mechanisms. (Expert Systems)

2.Implement BFS and DFS for solving a state-space problem and compare completeness, optimality, time and space requirements. (Blind Search)

3.Implement Greedy Best-First Search and A* for a path-finding problem and compare heuristic performance. (Heuristic Search)

4. Solve the 8-Puzzle Problem using A* and analyze nodes generated, nodes expanded, solution depth and solution cost. (State-Space)